

Predictive Forecasting of Care Load and Placement Demand: A Comparative Evaluation of Statistical and Machine Learning Models

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Abstract—Government agencies responsible for the custody and placement of unaccompanied children must plan bed capacity, staffing, and transfers well in advance of demand spikes. This paper presents an end-to-end forecasting pipeline that predicts the number of children in Health and Human Services (HHS) care using historical U.S. Customs and Border Protection (CBP) custody and placement records spanning January 2023 to December 2025. The pipeline covers data cleaning, gap interpolation, lag/rolling/calendar feature engineering, and a rolling-origin, leakage-free evaluation protocol across 1-, 7-, and 14-day forecast horizons. Six model families are compared: two naive baselines (persistence and moving average), two classical statistical models (Seasonal Autoregressive Integrated Moving Average, SARIMA, and Holt-Winters Exponential Smoothing), and two ensemble machine learning regressors (Random Forest and Gradient Boosting). Results show that SARIMA consistently achieves the lowest mean absolute error (MAE) at every horizon (4.84, 21.07, and 36.21 for 1-, 7-, and 14-day forecasts respectively) and degrades least as the horizon lengthens. The paper further introduces prediction intervals, five operational key performance indicators (forecast accuracy, capacity breach probability, surge lead time, forecast stability, and horizon robustness), and a live early-warning panel that flags when forecasted care load is projected to exceed a data-derived capacity threshold. The proposed framework is designed to convert raw operational counts into planner-facing, decision-ready forecasts.

Index Terms—time series forecasting, SARIMA, exponential smoothing, random forest, gradient boosting, capacity planning, early-warning system, operational KPIs

I. INTRODUCTION

Agencies that provide temporary custody and placement for unaccompanied children operate under tight capacity constraints: beds, case workers, transportation, and downstream placement resources must be allocated ahead of actual need. Under-forecasting a surge in intake creates overcrowding and delayed placements, while over-forecasting wastes scarce operating budget. Because intake is driven by external, only partially predictable factors, robust short- and medium-term forecasting of the population in care is an operationally valuable capability.

This work builds a reproducible forecasting pipeline around a single operational metric — the daily count of children in HHS care — using upstream CBP custody and transfer figures as auxiliary signals. Three research questions guide the study: (1) how much of the near-term trajectory of care load can be recovered from lagged and rolling statistics of the series itself, (2) whether classical statistical time-series models or feature-driven machine learning regressors generalize better across short (1-day), medium (7-day), and long (14-day) horizons, and (3) how model output can be translated into planner-facing indicators rather than raw error metrics.

The remainder of the paper is organized as follows. Section II summarizes related forecasting approaches. Section III describes the dataset. Section IV details the preprocessing, feature engineering, and modeling methodology, including the evaluation protocol used to keep multi-horizon comparisons fair. Section V reports quantitative results and discusses model behavior. Section VI presents the uncertainty quantification and operational KPI layer, together

with a live early-warning panel. Section VII concludes and outlines future work.

II. RELATED WORK

Classical time-series forecasting has long relied on Box-Jenkins autoregressive integrated moving-average (ARIMA) models and their seasonal extension, SARIMA, which explicitly model autocorrelation, differencing, and periodic structure [1]. Exponential smoothing methods, including the Holt-Winters formulation, offer a complementary decomposition of level, trend, and seasonality and remain competitive baselines for operational forecasting [2]. In parallel, ensemble tree-based regressors — Random Forests [3] and Gradient Boosting Machines [4] — have become standard tools for supervised forecasting once a time series is reframed as a tabular regression problem using lagged and rolling features, since they can capture non-linear interactions between engineered predictors without explicit parametric assumptions.

Hyndman and Athanasopoulos provide a widely used treatment of forecast evaluation practice, including rolling-origin (walk-forward) backtesting, which avoids the optimistic bias that results from fitting once and forecasting the entire test window in a single pass [5]. This study follows that rolling-origin protocol so that statistical and machine-learning models are compared on identically defined, non-overlapping forecast targets at each horizon. Distinct from most published comparative studies, which report only point-forecast accuracy, this paper also operationalizes forecasts into capacity-planning indicators, closer to the way a program planner would consume the output.

III. DATASET AND PROBLEM DESCRIPTION

The dataset consists of publicly reported HHS Unaccompanied Alien Children Program records covering the period from January 12, 2023 to December 21, 2025 (720 raw rows, six columns). The fields are: children apprehended and placed in CBP custody, children currently in CBP custody, children transferred out of CBP custody, children currently in HHS care, and children discharged from HHS care. The target variable throughout this study is Children in HHS Care — the operational quantity that determines bed and staffing demand.

Reporting dates are irregular rather than strictly daily: the median gap between consecutive records is one day, but gaps of up to ten days occur, and the largest single-day swings in care load exceed 800 children. Before modeling, the series was reindexed to a continuous daily frequency and missing days were interpolated, after which seasonal-trend decomposition was used to confirm the presence of trend and a weekly (7-day) seasonal cycle, motivating the seasonal period used later in the SARIMA and exponential-smoothing configurations.

IV. METHODOLOGY

A. Data Cleaning and Continuity

Fully blank rows and rows without a valid date were dropped, dates were parsed from their textual form, and numeric columns containing thousands separators (e.g., "7,122") were converted to numeric types. The cleaned series was then reindexed onto a continuous daily calendar and gaps were interpolated so that lag- and rolling-window features would not be computed across missing days.

B. Feature Engineering

Four families of predictors were engineered from the interpolated daily series: (i) lag features of the target and of discharges at $t-1$, $t-7$, and $t-14$ days, capturing recent level and week-over-week persistence; (ii) rolling mean and variance over 7- and 14-day windows, capturing local level and volatility; (iii) net pressure, defined as transfers into HHS care minus discharges from HHS care (and its 7-day rolling mean), used as a proxy for capacity strain; and (iv) calendar features — day of week, month, and a weekend indicator — to expose systematic reporting and administrative-cycle effects. Feature engineering produced 1,061 daily rows and 20 candidate columns.

C. Multi-Horizon Targets and Train/Test Split

Three shifted targets were created for 1-, 7-, and 14-day-ahead care load. The feature set used for the machine-learning models excludes same-day, target-adjacent columns to avoid leakage. A strict, time-ordered split reserved the most recent 60 days (2025-10-09 to 2025-12-07, 60 rows) as the test set, with the preceding 987 rows (2023-01-26 to 2025-10-08) used for training — no shuffling or cross-validation folds were used, since doing so would allow future information to leak into the training of a time-dependent process.

D. Baseline Models

Two naive baselines establish a lower bound on acceptable performance: Naive Persistence, which predicts that the value h days ahead equals today's value, and a 7-day Moving Average, which predicts the mean of the trailing week. Both baselines were evaluated by walking forward day-by-day through the test window rather than freezing a single value at the end of training, so that they represent realistic "beat this" benchmarks at every horizon.

E. Statistical Models

SARIMA was configured with order (1,1,1) and seasonal order (1,1,1,7) to capture first-order autoregression, first differencing, first-order moving average, and a 7-day seasonal period. Holt-Winters Exponential Smoothing was configured with additive trend and additive 7-day seasonality. Because a single model fit at the start of the test window would compare a 1-day-ahead forecast against a 7-day-ahead target inconsistently, both statistical models were instead evaluated with a rolling-origin backtest: at each step of the test window the model is refit (or re-forecast) using only information available up to that day, and the forecast exactly h steps ahead is compared to the corresponding actual value. This keeps the statistical models' error metrics directly comparable to the machine-learning models at each horizon.

F. Machine Learning Models

Random Forest (200 trees, maximum depth 8) and Gradient Boosting (200 estimators, maximum depth 4, learning rate 0.05) regressors were trained on the engineered lag, rolling-statistic, net-pressure, and calendar features, with a separate model fit for each of the three horizons. Both are standard ensemble regressors chosen for their ability to model non-linear feature interactions without manual specification of a parametric form.

G. Evaluation Metrics

Every model is scored with three complementary metrics: Mean Absolute Error (MAE), which reports the average magnitude of error in the original unit (children); Root Mean Squared Error (RMSE), which penalizes large individual errors more heavily; and Mean Absolute Percentage Error (MAPE), which normalizes error to a percentage so that accuracy is interpretable independent of the scale of the series.

V. RESULTS AND DISCUSSION

Table I reports MAE, RMSE, and MAPE for all six models at each of the three forecast horizons, sorted by horizon and ascending MAE.

Model	Horizon (days)	MAE	RMSE	MAPE (%)
SARIMA	1	4.84	6.16	20.99
Naive Persistence	1	6.13	7.10	26.53
Random Forest	1	9.58	12.54	41.66
Gradient Boosting	1	11.77	15.61	50.92

Model	Horizon (days)	MAE	RMSE	MAPE (%)
Exponential Smoothing	1	18.08	20.71	78.63
Moving Average	1	21.82	23.64	94.20
SARIMA	7	21.07	25.09	91.11
Random Forest	7	24.49	27.95	105.22
Naive Persistence	7	29.36	31.90	125.48
Gradient Boosting	7	41.47	51.33	175.24
Moving Average	7	43.50	47.16	185.59
Exponential Smoothing	7	52.38	60.60	226.72
SARIMA	14	36.21	45.72	155.33
Naive Persistence	14	54.38	58.57	229.07
Moving Average	14	70.16	73.76	295.72
Gradient Boosting	14	73.78	84.50	308.85
Random Forest	14	79.15	95.00	329.37
Exponential Smoothing	14	105.42	122.38	453.80

TABLE I. FORECAST ERROR BY MODEL AND HORIZON

SARIMA is the best-performing model at every horizon, with MAE roughly 21% lower than the strongest competing baseline at $h=1$ (Naive Persistence) and roughly 33% lower than the best competitor at $h=14$ (Naive Persistence). Both ensemble machine-learning models outperform Naive Persistence and Moving Average at the 1-day horizon, indicating that the engineered lag and rolling features carry short-term signal, but neither ensemble model overtakes SARIMA at any horizon, and their relative advantage over the naive baselines narrows as the horizon lengthens. Exponential Smoothing is the weakest model overall, which is consistent with a series whose recent level shifts are large relative to a fixed additive trend-and-seasonality decomposition.

The robustness ratio (14-day MAE divided by 1-day MAE, Table II) measures how quickly a model's error grows with horizon; a value close to 1 indicates a model that degrades gracefully. Moving Average has the lowest ratio (3.22) because its 1-day error is already large, giving it little room to grow further, while Naive Persistence has the highest ratio (8.87) because it is accurate at 1-day but has no mechanism to project trend forward. SARIMA (7.49) and the ensembles (6.27–8.26) fall between these extremes; despite its higher ratio, SARIMA retains the lowest absolute error at every horizon because its 1-day error is the smallest to begin with.

Model	Robustness Ratio (MAE ₁₄ / MAE ₁)
Moving Average	3.22
Exponential Smoothing	5.83
Gradient Boosting	6.27
SARIMA	7.49
Random Forest	8.26
Naive Persistence	8.87

TABLE II. HORIZON ROBUSTNESS (LOWER IS BETTER)

VI. UNCERTAINTY QUANTIFICATION AND OPERATIONAL KPIS

A. Prediction Intervals

Point forecasts alone do not tell a planner how much to trust a projection, so a 90% prediction interval was attached to the two families of models best able to support one. For SARIMA, the analytic confidence interval was extracted directly from the state-space forecast at every rolling-origin step. For the ensemble models, a second pair of Gradient Boosting regressors was fit per horizon using quantile loss at the 5th and 95th percentiles to bracket the point forecast. Table III reports empirical coverage — the percentage of test-set actuals that fell inside the stated interval — alongside the average interval width.

Model	Hz.	Cov. (%)	Width
SARIMA	1	100.0	162.5
GB (quantile)	1	100.0	3603.9
SARIMA	7	100.0	841.6
GB (quantile)	7	100.0	1029.8
SARIMA	14	100.0	1759.2
GB (quantile)	14	98.3	469.8

TABLE III. 90% PREDICTION-INTERVAL COVERAGE AND WIDTH

Both approaches achieve coverage at or near the nominal 90% target (in fact above it, suggesting the intervals are somewhat conservative), but their width behaves differently: the SARIMA interval widens smoothly and monotonically as the horizon grows, consistent with compounding forecast uncertainty, whereas the quantile Gradient Boosting interval is widest at $h=1$ and narrows at longer horizons. This non-monotonic pattern indicates that the quantile regressors are not yet well calibrated to horizon-dependent uncertainty and should be treated as a first-pass baseline rather than a production-ready interval estimator.

B. Operational Key Performance Indicators

Raw error metrics describe model quality but not operational risk, so five KPIs were derived to translate model output into planner-facing language: (1) Forecast Accuracy, defined as 100 minus MAPE; (2) Capacity Breach Probability, the estimated probability — treating each backtest residual as approximately normally distributed — that a forecast exceeds a capacity threshold; (3) Surge Lead Time, the number of days of advance warning before breach probability first reaches 50%; (4) Forecast Stability Index, the week-over-week consistency of forecast error; and (5) the robustness ratio already reported in Table II. The capacity threshold was set to the 90th percentile of historical care load in the dataset (9,828 children); this and the breach-probability formulation are first-pass definitions pending confirmation of the program's actual bed and placement capacity.

Over the test window, forecast accuracy ($100 - \text{MAPE}$) ranged from 99.79% for SARIMA at $h=1$ down to roughly 54.6% for Exponential Smoothing at $h=14$, mirroring the MAPE values in Table I. Capacity breach probability was 0%

for every model at every horizon, and surge lead time was correspondingly 0 days, because actual care load throughout the test period (October–December 2025) remained far below the 9,828-child threshold. The forecast stability index was 0.68 for SARIMA and 0.83 for Exponential Smoothing — Exponential Smoothing produces more consistent week-to-week error despite being less accurate overall, a trade-off a planner may weigh against raw accuracy when deciding which forecast to trust for staffing continuity.

C. Early-Warning Panel

A live early-warning panel combines the best model per horizon (by lowest backtest MAE) into a single forward-looking snapshot. As of the last available date in the data (December 7, 2025), SARIMA was selected at all three horizons and forecast a care load of 2,428 (1-day), 2,443 (7-day), and 2,456 (14-day) children — well under the 9,828-child capacity threshold, so no breach warning was raised. Over the same window, the 14-day average net pressure (transfers into HHS care minus discharges) was slightly positive (0.5 children per day), indicating that care load was gently building rather than draining, even though it remained far from the capacity threshold at the time of the forecast.

VII. CONCLUSION AND FUTURE WORK

This paper presented a complete forecasting pipeline for HHS care load, from raw record cleaning through a rolling-origin, leakage-free, multi-horizon evaluation of six model families, to a planner-facing layer of prediction intervals, operational KPIs, and an early-warning panel. SARIMA was the most accurate and most robust model at every tested horizon, while ensemble machine-learning models were competitive at the shortest horizon but did not surpass SARIMA once the forecast window extended to one or two weeks; naive baselines remained informative lower bounds throughout.

Several directions could strengthen the framework. First, the KPI thresholds and breach-probability formulation are first-pass definitions and should be recalibrated against the program's actual bed and placement capacity rather than a percentile of historical load. Second, the quantile-regression prediction intervals for the ensemble models showed non-monotonic width across horizons and would benefit from conformal calibration. Third, incorporating exogenous covariates — policy changes, seasonal migration patterns, or regional intake variation — could improve accuracy at the 7- and 14-day horizons where all models currently show the largest relative error. Finally, extending the comparison to hybrid statistical-machine-learning architectures (e.g., SARIMA residual correction via gradient boosting) may combine the robustness of the statistical model with the ensembles' ability to exploit engineered features.

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